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The orchestrated simulation method (OSM) for virtual simulation in incident command training☆

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ABSTRACT

Virtual Simulation (VS) using computer simulation technologies offer promising opportunities for education and training of emergency professionals. However, the introduction of VS into existing training practices involves challenges that extend beyond technology investment. Organizations need to adjust training practices and develop new competencies for which structured guidance is often lacking. This paper presents the Orchestrated Simulation Method (OSM), a structured practice-based method developed to support Fire and Rescue Services (FRS) organizations in introducing, delivering, and continuously improving VS for Incident Command (IC) training. OSM was developed through action research across eight iterations in four representative Swedish FRS organizations, involving 27 instructors and two VS experts who prepared and conducted training for 418 learners at different IC levels. The focus is on the instructor team's introduction, management and refinement of VS training, rather than learner performance. The findings suggest that OSM supports the alignment of learning objectives, instructor roles, scenario design, assessment practices, and organizational conditions, thereby helping FRS organizations and instructor teams introduce and manage VS across different IC training contexts. However, further research is needed to assess generalizability.

1. Introduction

Practice-based training in simulated scenarios plays a central role in developing emergency management competence (WHO, 2017). For Fire and Rescue Service (FRS) Incident Command (IC), simulation is commonly delivered through Live Simulation (LS), picture-or video-based simulations, or, increasingly, Virtual Simulation (VS). LS, conducted on physical training grounds using real equipment and live fire, has long been seen as the near-real training format. While simulating training scenarios using pictures or videos offers less resource-intensive alternatives, they provide fewer opportunities for applied learning in realistic operational situations (Hammar Wijkmark et al., 2019). In recent decades, advances in technology have enabled the development of VS, a simulation-based training format in which technology is used to provide computer-generated interactive three-dimensional learning environments, for example, Virtual Reality technology (VR). These environments serve as virtual training grounds, where users (e.g., learners and instructors) can assume roles, interact with others, and perform tasks relevant to the simulated situation and learning objectives. VS has gained attention for its promise of safe, flexible, repeatable, and potentially cost-efficient training solutions for applied learning and development of IC competencies such as situation awareness and decision-making (Lamb, 2014; Polikarpus et al., 2019; Reis & Neves, 2019; Williams-Bell et al., 2015).

However, the successful introduction and use of VS as a new training format in an organization requires more than acquiring the technology. Organizations must develop pedagogical strategies, instructor competence, and organizational routines to effectively integrate technology into their training practices (Fomin et al., 2024; Theodorio, 2024; Verkuyl et al., 2024). For organizations with limited experience of VS, the introduction can be complex and resource-intensive, characterized by unclear goals and uncertainties regarding how to align VS with organizational needs and pedagogical practices (Kozlowski et al., 2016), which is essential for achieving intended learning objectives (Polikarpus et al., 2023). Previous research has shown that organizations often struggle to adopt and sustain training using VR due to missing technical,

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organizational, or instructor readiness (Hancko et al., 2025), and lack of instructor support in planning, delivering, and continuously improving the training (e.g., Engelbrecht et al., 2019). While selecting appropriate technologies is important, the wide range of ways they can be applied in training activities introduces additional complexity to their seamless use. In practice, the instructors must align VS training with organizational conditions, understand the needs and capabilities of technologies, design goal-oriented learning scenarios, coordinate instructor teams, manage multiple roles, structure assessments, and support systematic improvement over time.

While several frameworks exist to support different aspects of VS and VR integration, they primarily focus on technology acceptance, learners' outcomes, or broader pedagogical integration, as we argue in Sections 2.3 and 6.1. They offer limited support for the practical introduction, delivery, and continuous improvement of VS. In particular, there remains a lack of structured guidance for instructor teams responsible for integration, management, and continuous improvement of VS in resource-constrained contexts.

This paper presents the Orchestrated Simulation Method (OSM), a structured and practice-oriented method developed to support FRS instructors and organizations in introducing, delivering, and improving VS in IC training. OSM focuses on developing capability in organizations with limited prior experience with VR technologies and VS.

OSM conceptualizes the introduction and use of VS as an orchestrated process that emphasizes the alignment of learning objectives, instructor roles, scenario design, delivery, assessment, and systematic improvements while considering organizational constraints and resources. OSM was developed and evaluated following the Action Research (AR) method (Baskerville, 1999), conducted between 2020 and 2024 across four Swedish organizations responsible for IC education and training: the government agency responsible for IC education and three regional FRS organizations providing annual training development for their commanders. Across eight iterations (two at each organization), 27 instructors (individuals) and two VS experts prepared and delivered training for a total of 418 commanders (learners) of different levels. AR (Baskerville, 1999) provided the iterative mechanism through which OSM was tested and refined in authentic training settings.

The eight iterations (using OSM in eight IC educational modules) were chosen to support representative introduction, instructor learning, and development of sustainable, organizational use of VS, while minimizing reliance on external technical support. In this paper, sustainable use refers to an organization's capacity to introduce and manage VS as a stable, yet continuously evolving part of training practice. This includes maintaining technical functionality, developing instructor competence, and continuously adapting and refining training across different learning objectives, contexts, technologies, and command levels. To support instructors' understanding of how to plan, deliver, and coordinate training for diverse learners, OSM is described through an orchestration metaphor that emphasizes alignment of roles, timing, coordination, and iterative refinement of practice. The metaphor offers a practical lens for managing the complexity of VS in IC training.

The paper is organized as follows: Section 2 covers the background and theoretical foundations, Section 3 outlines the research methodology, Section 4 introduces OSM, and Section 5 details the application of OSM in practical settings and presents results from its use. Section 6 discusses OSM in relation to other frameworks and methods introduced in Section 2 and presents current limitations and future work. Section 7 provides the conclusions.

2. Background

This Section presents the practical and theoretical foundations of this research, beginning with the specific context of IC training in Sweden, the role of VS within IC training, existing implementation approaches and the need for structured support, and ending with an enumeration of theories and methods currently used.

2.1. Briefly on incident command (IC) training

Incident commanders operate in high-stakes environments characterized by uncertainty, time pressure, and rapidly evolving conditions (Butler et al., 2024; McLennan et al., 2024). FRS command systems are commonly structured across command levels that reflect increasing operational and organizational responsibilities, although the terminology, responsibilities, and organizational arrangement differ between countries; for example, in the UK (Hanlon, 2008), USA (National Incident Management System | FEMA.gov., 2025), and Sweden (ELS, 2024): Despite the differences, higher command levels generally require increasingly complex training, influencing resource requirements, scenario, scenario design and complexity, instructor team composition and roles, and the design of training activities.

The command levels examined in this research follow the Swedish FRS command structure as described in the national handbook ELS (2024) focusing on the operational and tactical command (IC1, IC2, IC3) which represent progressively increasing levels of responsibilities: a commander at level IC1 is responsible for leading a single firefighter team (typically up to five firefighters), whereas commanders at IC2 and IC3 coordinate multiple units and broader incident response activities. While the command system and associated roles are defined in ELS, the educational requirements and learning objectives for these levels are specified through national curricula and training programs administered by the Swedish Civil Defence and Resilience Agency (MCF) and regional FRS organizations. However, these frameworks provide limited guidance regarding how instructor teams should introduce, use, organize, and manage VS in training practices. Effective command performance requires the development and maintenance of both technical and non-technical competencies across a wide range of operational contexts, from routine incidents to rare and complex emergencies. In addition, the increasing responsibilities associated with different command levels place different demands on training, requiring variation in scenario complexity, available resources, instructor support, and opportunities for practice. Exposure to training alone is not sufficient to systematically acquire the necessary skills, rules, concepts, or attitudes that result in improved work performance (McLennan et al., 2024). Training activities need to be designed to support learning progression and the achievement of clearly defined learning objectives, focusing on technical knowledge but also on non-technical skills such as situational awareness, communication, leadership, and decision-making (Flin et al., 2008; Lamb et al., 2021).

Traditionally, IC training combines classroom-based simulation and LS. Classroom approaches that use photos, videos, or physical models are resource-efficient but offer limited opportunities for applied, context-rich non-technical skills training in dynamic scenarios. LS provides greater realism through scenarios staged on a training ground, with live fire and real equipment, but is constrained by cost, safety considerations, environmental regulations, and limited opportunities for repetition and variation (Argyris, 1977; Hammar Wijkmark & Heldal, 2020). At the same time, emergency response contexts are becoming increasingly complex due to evolving risks, technologies, new building materials and vehicles, and organizational structures (Alexander, 2003; Cohen et al., 2013; Engelbrecht et al., 2019; Hancko et al., 2025). These developments increase the need for training approaches that are flexible, scalable, and capable of supporting advanced cognitive and collaborative competencies. Designing such training requires careful pedagogical planning. Approaches such as experiential learning (Kolb, 2014), recognition-primed decision-making (Klein, 1997), and double-loop learning (Argyris, 1977) highlight the importance of structuring learning experiences that support reflection, adaptation, and progression.

2.2. Virtual simulation using virtual reality

Throughout this paper, we refer to VR as computer-generated,

interactive, three-dimensional models that allow users to act according to its implemented rules. VS refers to the simulation-based training format in which VR and other technologies are used to enable learners to perform in their professional roles, where instructors facilitate scenarios and assess performance (e.g., Hammar Wijkmark et al., 2019; Polikarpus et al., 2021; Reis & Neves, 2019).

In contrast to many studies that focus on fully immersive VR using head-mounted displays (HMDs), e.g., Radianti et al. (2020) and Zhang et al. (2025), this research focuses on non-immersive configurations, using large screens. For IC training, non-immersive VR is often preferred because it allows learners to use their work-related tools, such as radios, notebooks, and command-support systems, while maintaining awareness of multiple information sources. Fully immersive solutions may restrict these activities and are therefore less suited for a complex decision-making context (Hammar Wijkmark, 2024; Kapalo et al., 2022).

While VR promises opportunities for safe, repeatable, and context-rich training, its effectiveness depends on the instructional design rather than technology alone. As emphasized by Salas et al. (2012), technology alone does not produce learning; learning depends on the instructional features embedded in and surrounding its use. In this paper, effective use of VR in training therefore means embedding it in a VS training activity, including carefully aligned scenario design, facilitation where instructors can manage learners' actions, interactions, assessments, and introduce information at appropriate times. These elements must be consciously managed and synchronized to enhance learner engagement, promote a sense of presence, support psychological safety, allow exploration and "safe failure", and create opportunities for reflection and progression (Brodin et al., 2026; Fanning & Gaba, 2007; Gaba, 2007; Issenberg et al., 2005; Radianti et al., 2020).

Training in VS can be delivered either on-site or remotely. In on-site VS, instructors and learners are co-located, allowing face-to-face role-play interactions to complement the virtual environment. In remote VS, instructors and learners are geographically separated (e.g., across rooms, buildings, or further distances), and roleplay is conducted through mediated channels, using avatars and voice communication to support training (Hammar Wijkmark et al., 2019). Both on-site VS and remote VS have the promise to support collaboration with others and allow to perceive adequate sense of being present, commonly described as "being and acting in the same situation" together with other participants (Hammar Wijkmark et al., 2019; Schroeder et al., 2001). Compared to on-site VS, remote VS increases instructional complexity, requires a coordinated team of instructors, additional technical and collaborative competencies for coordination, and clear instructional roles. All iterations examined in this research used a remote VS configuration.

2.3. Challenges for considering virtual simulation in education

Simulation-based education has long been recognized for its potential to support practice-based training. Its pedagogical promise lies in creating realistic, goal-oriented learning settings in which learners can act, experience consequences, and reflect in a safe, non-invasive environment. Simulation-based IC training is often grounded in influential learning theories, such as recognition-primed learning (Klein, 1997) and Kolb's (2014) experiential learning theory. These emphasize cycles of concrete experience, reflection, conceptualization, and experimentation, which align naturally with scenario-based training. Learning objectives and progression in training and education are often structured using Bloom's taxonomy (Bloom et al., 1956), guiding learners from foundational understanding of the problem leading to knowledge creation. To support such learning, scenarios must enable learners to engage in increasing levels of complexity while remaining responsive to unfolding decisions and actions. However, traditional scenario design is often organized around predefined timelines, scripted events, and predetermined injects (i.e., cues that guide learner attention), which may limit adaptability and reduce decision realism (Tena-Chollet et al., 2017).

Achieving this balance places significant demands on instructors responsible for planning and facilitating VS. VS configurations require that they serve several simultaneous roles, for example facilitator, operator, role player, communication coordinator, and assessor. These roles require a combination of technical fluency, pedagogical judgment, and the ability to orchestrate coordinated team performance (Alklind Taylor, 2014; Van Laere et al., 2021; Verkuyl et al., 2024). Recent studies show how mixed reality and training are sustained through continuous participant work to manage breakdowns, uncertainties, and technical limitations. Instructors and learners need to actively consider workarounds, adaptations, and improvisations to keep the simulation going, which can involve shifting roles, meanings, and interactional frames (Sjöberg et al., 2025). While such adjustments help maintain activity and engagement, they may also introduce a gradual shift away from intended learning objectives toward maintaining the perceived realism or technical continuity. This requires a deeper understanding of simulation design and delivery, where increasing representational or interaction fidelity does not necessarily lead to improved learning outcomes (Alessi, 1988; Bonfert et al., 2025).

Despite its pedagogical potential, the introduction and management of VS as a training modality remain challenging in relation to maintaining alignment between learning objectives, scenario design, and instructional facilitation (Polikarpus et al., 2021; Radianti et al., 2020) while also considering organizational preconditions such as dedicated instructor time and scheduled simulation sessions (Fomin et al., 2024).

Several frameworks and models have been developed to support adoption or use of technology, also for VS and VR. These include general models such as the Technology Acceptance Model (Davis, 1989), the Kirkpatrick evaluation model (Kirkpatrick & Kirkpatrick, 2009), the Technological Pedagogical Content Knowledge (TPACK) framework (Bull & Bell, 2009), the Situated Cognition Model (Lui & Goel, 2022) connecting the learning effectiveness of technology to the different contexts. The TRIFECTA concept (De Vaujany et al., 2018) explicitly argues for alignment among organizational rules, practices, and the technology (Fomin et al., 2024), not only for training but also to help adopters anticipate organizational friction (e.g., IT policies, cultural resistance). These models require interpretation for each training context and do not provide operational guidance for planning, coordination, delivery, or iterative improvement of VS training. They focus more on understanding key issues related to the training than on managing them in practice. Other approaches focus on learning through immersive VR, emphasizing simulation-specific characteristics such as presence, cognitive load, and emotional engagement, e.g., CAMIL (Makransky & Petersen, 2021), making causal pathways explicit, and diagnosing which design components lead to better learning. Another example of this category is FIVE (Slater & Wilbur, 1997), which distinguishes immersion (a technological property of the graphical environment) from the user's inclusive, surrounding experience (presence). While the commanders' learning and presence are important for this research, the primary focus is on how instructors can create conditions that enable such learning. The more practice-oriented frameworks, such as the Simulation-Based Learning Environment (SBLE) model (Keskitalo & Ruokamo, 2015), and the Collaborative Authoring Process Model (CAPM) (Polikarpus et al., 2021), focus on specific aspects of instructional design, scenario development, and facilitation. While all of these frameworks provide valuable insights, they tend to focus on specific aspects, such as technology acceptance, pedagogical design, or learning objectives. They provide limited guidance on how instructors can manage the full lifecycle of VS, i.e., including initial introduction, instructor capacity building, planning, delivery, assessment, refinement, and sustained organizational use (see further Table 3's result from Section 6.1).

As a result, FRS organizations may remain dependent on external expertise for introduction and management (Hammar Wijkmark, 2024)

or struggle to move beyond the initial implementation and basic use of VS (Polikarpus et al., 2023). This highlights the need for a structured, instructor-centered approach that supports VS as an integrated, continuously evolving training format. OSM is proposed as a method that supports instructors at FRS organizations with limited or no prior VS experience in introducing, developing, and sustaining VS while ensuring pedagogical alignment, operational realism, and long-term organizational capability.

3. Methodology

This research employed Action Research (AR) as outlined by Baskerville (1999), which combines organizational problem-solving with the development of scientific knowledge through iterative cycles of interventions and reflections. AR was selected because the aim of this research was to produce research-based, actionable knowledge for practitioners and to contribute to a better understanding of how VS can be introduced, used, and sustained in training practice. AR requires close collaboration between researchers and practitioners and emphasizes the development of solutions grounded in practical needs and relevant theories. Following Baskerville (1999), each iteration involved diagnosis, action planning, action taking (intervention), evaluation, and reflection, with researchers and practitioners actively participating throughout the change process.

While Baskerville (1999) outlines the procedural logic guiding each iterative development of OSM, quality criteria, defined by Bradbury and Reason (2008) on rigor, participation, and practical relevance of AR guided the involvement of researchers and practitioners, and the selection of participating organizations and iterations (presented in Table 1) with focus on; relational quality, ensuring genuine collaboration and shared decision making between researchers and practitioners; practical quality, ensuring that the interventions addressed real organizational needs and contributed to meaningful improvements in local practice; and epistemic quality, ensuring that the inquiry generated trustworthy, theoretically grounded, and practically relevant knowledge.

Table 1
Overview of the iterations (I#) involving separate instructor groups at the organizations involved, the time period the activities were followed, the training level and focus, and the aims to use VS[a] for each selected iteration.

I#	Organization/ Time	Training level/Focus	Aim to use VS
I1	MCF1/Summer 2020	IC1: Final exam within the national IC1 education, demonstrating the necessary competencies for IC1.	Enable remote final examination using VS during Covid-19 pandemic
I2	Skaraborg1/ Spring 2021	IC1: Annual development training.	Develop a simulation center, supplement LS, and increase instructor competency
I3	MCF2/Autumn 2021	IC1: Basic training and examination exercises within the national IC1 education.	Integrate VS in the new national IC1 education to be launched in January 2022
I4	Skaraborg2/ Spring 2022	IC2, IC3: Annual development training.	Enable advanced IC2 and IC3 development using VS
I5	Nerike1/ Autumn 2022	IC2, IC3: Annual development training.	Evaluate VS for annual development training
I6	Karlstad1/ Autumn 2023	IC1, IC2, IC3: Annual development training.	Evaluate VS for annual development training
I7	Nerike2/ Autumn 2023	IC1: Annual development training.	Further IC1 development based on VS after prior experience
I8	Karlstad2/ Autumn 2024	IC1, IC2, IC3: Annual development training.	Further IC1–IC3 development based on VS after prior experience

[a]Remote VS was used in the studied iterations.

3.1. Involving practitioners and researchers

This research involves two authors with practical experience in FRS training, introduction, and use of VS. The first author (practitioner-researcher) has more than 18 years of experience organizing and conducting VS training and exercises across multiple emergency management organizations. The third author (practitioner-educator) has 30 years of experience serving as an incident commander at different command levels and more than 16 years of experience in IC education and training, and contributed to the development of data collection procedures, the refinement of OSM and its terminology. These authors' prior practical experiences indicated a need for more structured instructor support when introducing and managing VS. This observation provided the initial motivation for developing the OSM. The second author contributed contextual understanding of the problem and anchored it in existing research.

The participating organizations initially contacted the authors with practitioner background because of their expertise in VS. Due to the increased demand for such training, they realized the benefit of defining a systematic approach to support organizations in introducing, organizing, and sustaining VS within their IC training context. Across the eight iterations, the researchers and practitioners collaborated closely with the participating organizations by supporting the planning activities, scenario development, instructor preparation, training delivery, evaluation, and reflection.

3.2. Research contexts and participating organizations

The examined contexts consist of organizations responsible for IC education and training in Sweden, focusing on instructor teams introducing and using VS. Four organizations participated: MCF, the Swedish government agency responsible for the national IC education program and three regional FRS associations that provide recurrent training and professional development of employed commanders. The national program delivers foundational education and examinations within the mandatory IC1 qualification, whereas the regional FRS associations provide recurrent training and development for personnel at IC1, IC2, and IC3 levels. During the period examined, VS was introduced for IC1 at all participating organizations, while VS for IC2 and IC3 was implemented exclusively at the regional FRS organizations.

The selected organizations represent the diversity of Swedish IC training practice: the MCF College Sandö and the regional FRS associations of Skaraborg (19 FRSs), Nerike (9 FRSs), and Karlstad (6 FRSs). Their structures reflect typical Swedish FRS organizations and enable the investigation of VS introduction and use across both formal IC1 education (2 IC1 classes at MCF) and within recurrent professional development contexts (6 commander groups at FRSs), addressing varying command levels, learning objectives, and organizational conditions.

3.3. Iterative development and training contexts

Each of the eight iterations (I1-I8) represented a distinct training context involving either an IC class at MCF or a group of employed commanders within a regional FRS organization. The iterations varied in organizational context, purpose, and training objective depending on the targeted command level. All iterations followed the AR cycle outlined by Baskerville (1999): **Diagnosis:** analysis of the local context, training objectives, and instructor team preconditions. **Action planning:** design and preparation of training activity for the specific context. **Action taking:** delivery of VS training. **Evaluation:** assessment of training activities, use of VS, instructor performance in delivery, feedback and assessment practices. **Reflection:** organizational learning and refinement of training practices and OSM.

Table 1 summarizes the iterations and their purposes for introducing VS, and organizational conditions while enabling refinement of OSM.

Additional details on instructor teams, technical setup, training scope, scenario complexity, and preparation time are provided in Appendix A.

3.4. Technology used

All iterations used the commercially available XVR On Scene virtual simulation platform[1,2], which provides a three-dimensional environment for building and conducting interactive emergency management scenarios. The platform operates on standard computers and screens, speakers, and game controllers. Although immersion through HMDs was technically possible, participating instructors concluded that HMDs conflicted with typical IC work practices, such as radio communication, note-taking, and the use of digital command tools. Consequently, all organizations employed a non-immersive remote VS configuration combining large screens with the tools normally used by commanders in operational practice. This configuration enabled VS to be integrated with the organizations' established command procedures while maintaining accessibility and minimizing technical complexity.

3.5. Participants and empirical material

Across the eight iterations, 27 instructors prepared to deliver training for a total of 418 learners (during the corresponding years). All participants provided informed consent for participation and data collection. No personal or operationally sensitive information was collected. The empirical material also included documentation from scenario development, planning, and training sessions, as well as iterative reflection activities. Throughout the research period, researchers and practitioners jointly documented challenges, improvement needs, unsuccessful experiences, and successful outcomes. These observations were continuously discussed during reflection activities and retained in the analysis (see Appendix B and Section 5), and used to support iterative refinement of OSM (Section 4).

4. The orchestrated simulation method (OSM)

This Section presents the OSM, and how it operationalizes the introduction and use of VS training in organizational practice using seven phases.

4.1. Motivations to define OSM

For organizations unfamiliar with VR and VS, the introduction of such technology is often perceived as a challenge primarily focused on acquiring and operating the technology. OSM was conceptualized by the two practitioner authors through an orchestration metaphor, emphasizing coordinated roles, timing, rehearsal, and iterative refinement necessary for planning training. Similar to preparing a musical ensemble, effective VS training depends on aligning multiple interdependent components within the organizational context, including rules and regulations that involve others (Fomin et al., 2024).

Basic issues to be considered at the start were the systematic alignment between the different activities and learning goals (Biggs, 1999), considering experiential learning (Kolb, 1984), especially for simulation and games (Backlund et al., 2013) and considering Bloom's taxonomy (Bloom, 1984; Bloom et al., 1956) for learning. While there are several methods and frameworks for VS introduction (as we argued in Section 2.3), with pedagogy and learning in focus, e.g., via TPACK (Bull & Bell, 2009) or CAMIL (Makransky & Petersen, 2021), none of these methods and frameworks include explicitly information on contextual VS support for instructors in independently planning training. Therefore, in close collaboration with practice and motivated to define support for instructors facilitating VS-training use and management, the seven phases of OSM were defined (see Fig. 1), used, and refined during each of the eight AR iterations.

OSM's seven phases outline activities to focus on, ensure the goals are delivered in each phase, and provide a checklist to support instructors in facilitating continuous planning through all phases. Phases 1–4 focus on preparation, including learning objectives, organizational prerequisites, scenario design, and instructor team development. Phases 5 and 6 address training delivery, feedback, and assessment, while Phase 7 focuses on evaluation, systematic updating, and continuous improvement over time. Outputs from each phase serve as input to subsequent phases, while experiences from training delivery and evaluation are fed back into earlier phases to support iterative refinement, continuous development, and sustainable use of VS.

The four organizations and contexts were chosen based on their articulated need to investigate the feasibility of achieving VS training by fulfilling the requirements of Bradbury and Reason (2008). The relational quality was ensured by considering natural collaboration and decision-making situations and by anchoring OSM in the existing, practical education. The practical quality was addressed by working with organizations that seek to improve their local practice and allow them to use VS in the near future, with the ambition of performing VS training without VS-expert help or other technical support. The epistemic quality was supported by considering systematic (helped by goals and checklists) decisions about necessary data, interpretation, and actions that are well-reasoned, transparent, and methodologically sound at each organization involved.

The first iteration focused on structuring initial use under limited VR and VS experience, while later iterations focused on support for refinement, transferability, and increased organizational autonomy. This progression allowed OSM to be shaped and examined through practical use across varying pedagogical, technical, and organizational conditions. Phases 1, 2, and 3 should involve only selected instructors responsible for developing VS capability and who hold pedagogical and technical responsibilities, while the entire instructor team should be involved from Phase 4 onwards.

To support practical application, each phase includes clearly defined deliverables (goals) and a structured checklist (Appendix C) that translate the phases into concrete activities, decisions, and considerations for instructors and organizations. The checklist operationalizes the phases by identifying key tasks, considerations, and decisions helping instructors ensure that critical elements are not overlooked. Consistent with the orchestra metaphor, the phases reflect stages involved in preparing and performing a musical piece, from selecting the repertoire and preparing the stage, to rehearsal, performance, and continuous refinement. Together, the phases and checklists may provide an intuitive structure for understanding how individual activities contribute to the overall VS training practice, while maintaining coherence throughout the process from introduction to VS management.

4.2. OSM Phase 1 — define learning objectives and assessment criteria

The purpose of Phase 1 is to translate formal curriculum requirements, regulatory standards, and organizational competence frameworks into a set of observable and actionable learning objectives and assessment criteria that can be achieved and evaluated through a simulation activity. Learning goals, objectives, and assessment criteria may already exist, be open for revision, or need to be developed from the outset. Regardless of their origin, they should be clearly documented and understood by all instructors involved.

To support and fully utilize the applied nature of VS, the learning objectives are primarily formulated at the third level of Bloom's taxonomy (Bloom, 1984; Bloom et al., 1956), the "apply" level. At this level, learners are expected to demonstrate knowledge through action in context, enabling the observation of both technical and non-technical competencies during scenario execution. Learning objectives focused

[1] https://www.mcf.se/en/.
[2] https://learnprogroup.com/xvr/.

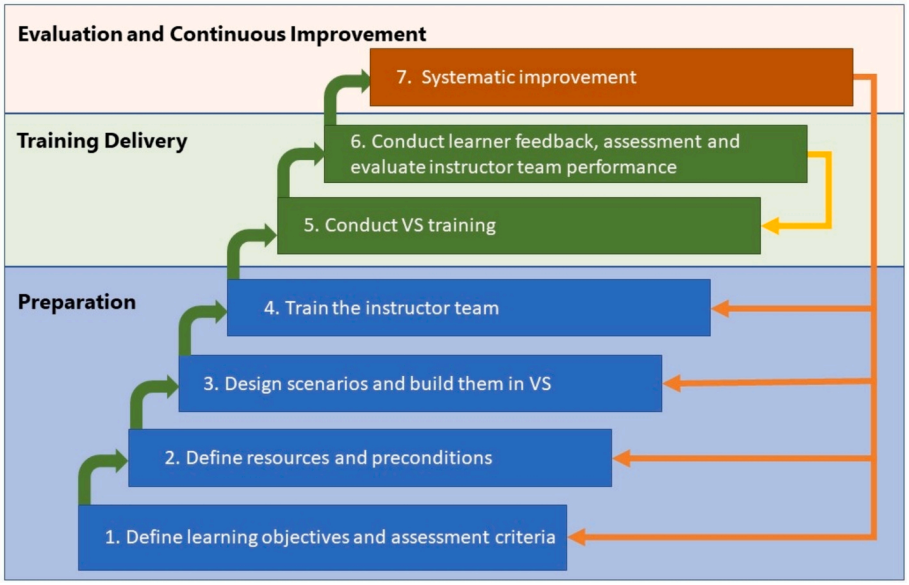


Fig. 1. The seven phases of OSM. Each phase delivers outputs to subsequent phases (green arrows), while experiences from delivered training and evaluation are fed back into earlier phases (orange arrows) to support continuous improvement. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

solely on lower cognitive levels, such as remembering, describing, or explaining concepts, can often be assessed through written exercises or classroom-based activities and therefore do not necessarily motivate a simulation-based learning environment.

Applied-level objectives enable the capture of performance and behavior that can be meaningfully enacted while the learner performs in the assigned role within a scenario.

Each learning objective is formulated using measurable verbs accompanied by a set of observable indicators and assessment criteria that define how the achievement of the objectives can be recognized during scenario enactment. Many organizations already employ standardized assessment or feedback frameworks, such as structured feedback methods, grading rubrics, behavioral marker systems, or examination protocols. These are considered in this phase to align with the evaluation criteria, ensuring methodological continuity and supporting instructor uptake. The resulting criteria specify what constitutes acceptable performance and provide a basis for consistent assessment across instructors and scenarios. The learning objectives and associated assessment criteria are subsequently reviewed and calibrated in collaboration with subject-matter experts. This work may draw on prior work described by Polikarpus et al. (2019) or the established Effective Command framework (Effective Command, 2025). Review and calibration may ensure that the objectives are relevant for the targeted command level (e.g., IC1, IC2, or IC3) and that the assessment indicators correspond to observable behaviors in real-world practice, and that assessors maintain a common interpretation of performance criteria through periodic review activities (Lamb et al., 2020).

The main deliverables of Phase 1 are:

- Clearly defined learning objectives for the specific training activity.
- Aligned assessment criteria specifying expected levels of performance.
- A learning-objective evidence map that (a) describes observable behaviors and actions linked to each learning objective, and (b) serves as a structured template for collecting performance evidence during scenario execution.

These deliverables establish the foundation for the subsequent phases of OSM. In particular, they inform the capacity requirements and scenario design in Phases 2 and 3 and later, provide the basis for the performance evaluation and learning assessment in Phases 5 and 6.

4.3. OSM Phase 2 — define resources and preconditions

The purpose of Phase 2 is to identify and establish the organizational, temporal, and technical resources and preconditions for reliable and sustainable use of VS. While Phase 1 defines what should be learned and assessed, Phase 2 establishes the practical conditions under which the training can be delivered.

The phase begins by identifying available personnel, their prior experience with VS or LS, and their readiness to participate in the planned training activities. Adequate staffing must be ensured for all required roles, such as operators, role-players, and assessors. The operator role often demands greater technical familiarity and interest, potentially creating a bottleneck for implementation capacity and long-term sustainability of VS use (Fomin et al., 2024; Radianti et al., 2020). Decisions regarding the delivery format should also be made during this phase, including whether the sessions will be conducted on-site or remotely, as well as the location, size, and other required characteristics supporting the technical setup.

Training schedules are established for both the initial introduction of VS and its planned recurrent use. During first-time use, sufficient time must be allocated for instructor training, familiarization with the simulation platform, and rehearsal of training procedures. The training session schedule should not just be planned for training delivery, but also include protected instructor time for technical setup or checks, participant briefing, and feedback activities. In addition, participant group size and training configuration must be determined, including

whether training will be conducted individually, with peer observations, or in other group constellations. Together, these factors determine the time available and required for each scenario and influence the complexity and scope of the design. For example, in I3 (Appendix A), training in VS was scheduled over two days for a class of 30 learners in groups of six. This constraint limited the duration of each scenario to approximately 15–20 min and required six separate scenarios to ensure sufficient training opportunities. Furthermore, because the training took place during the third week of the IC program, scenarios of relatively low complexity were required.

In this phase, the simulation platform has often already been selected and procured; if not, procurement must be conducted before further development can proceed. Technical preparation includes configuring workstations, audio equipment, communication channels, control interfaces, and network connectivity, followed by structured test sessions. These test sessions verify that the technical environment is stable, predictable, and aligned with pedagogical intentions.

The learner pre-briefing format needs to be established. Depending on available resources, schedules, and locations, the pre-briefing may be conducted in person, via a video call, or an instructional video, individually or through a group session. Consistency between pre-briefing content, learning objectives, and technical environment is essential for learner orientation and safety (e.g., Makransky & Lilleholt, 2018). The pre-briefing clarifies the purpose of the training, how to use the technical environment, how performance will be assessed, and how feedback will be provided. By reducing uncertainty regarding the training situation, pre-briefing supports psychological safety, i.e., a belief that the team (training situation) is safe for risk-taking (Edmondson, 1999). Recommended pre-briefing elements include: introduction of instructors and clarification of their roles during the session, description of the training format and communication structure, explanation of required technical interactions (e.g., interface elements, system affordances and limitations), and an introductory familiarization scenario in which the learner can practice navigation and interaction within the environment before participation in assessed training activities.

Phase 2 also includes preparing supporting documentation, such as feedback and assessment templates, communication protocols (e.g., radio channels if radios are to be used in the training setup), technical pre-training checklists, and learner pre-brief materials (e.g., videos of documents). These artifacts promote consistency across instructors and training sessions, establish shared expectations, and support reliability over time.

The main deliverables of Phase 2 are:

- Staffing plan.
- Training schedule with protected instructor preparation time.
- Technical setup and readiness checklist.
- Communication procedures, channels, and supporting technology.
- Learner pre-briefing materials.

Phase 2 establishes the organizational and technical foundations for subsequent phases of OSM. In particular, the deliverables define the resource constraints and operational conditions that guide scenario development in Phase 3 and support reliable preparation, execution, feedback, and assessment in Phases 4–6.

4.4. OSM Phase 3 — design scenarios and build them in VS

The purpose of Phase 3 is to design, structure, and technically build the virtual scenarios that operationalize the learning objectives defined in Phase 1, while accounting for the organizational and technical constraints identified in Phase 2.

Scenario development begins with collaborative planning (e.g., paper-based), in which instructors identify critical events, decision points, role allocations, and information flows required to elicit the targeted learning objectives and to make assessment criteria observable and measurable. Approaches such as the CAPM (Polikarpus et al., 2021) may support this process. The scope and complexity of the scenario influence the number and variety of roles involved, including victims, bystanders, agency representatives, and command roles. Particular attention is given to those roles that are directly influencing the learner's decision-making process. For example, in an IC2 scenario, key interactions may involve subordinate commanders (IC1), or commanders from collaborative agencies (e.g., the police).

Scenario planning includes specifying role responsibilities, available information, operational objectives, and behavioral intentions. Rather than prescribing rigid scripts, the design focuses on providing sufficient guidance for instructors to portray roles consistently and realistically while retaining flexibility to respond to the learners' actions. Semi-scripted role-play interactions and operator-triggered events are drafted to support the intended pedagogical trajectory and may be represented through avatars or other entities in the virtual environment.

Once the conceptual design is complete, the scenario is implemented in the virtual simulation platforms. This includes configuring the physical and societal environment as well as creating dynamic elements such as fire development, avatar and crowd movements, environmental changes, communication events, or other operational events and triggers.

Supporting materials such as operator guides and role-play instructions are documented to ensure consistency across instructors and repeated training sessions (e.g., when new instructors are included or after longer breaks).

Each scenario undergoes both technical and pedagogical testing. This verifies system functionality, scenario logic, pacing, realism, and alignment with the learning objectives, assessment criteria, and available scheduled time and instructor resources.

Within the OSM, scenario design is organized around operational episodes rather than a fixed chronological timeline (Keskitalo & Ruokamo, 2015). Each episode represents a meaningful operational situation in which the learner is required to observe, assess, decide, and act (related to Klein, 1997; Kolb, 2014). This approach, together with the flexible role descriptions, enables instructors to orchestrate the intended challenges and learning opportunities by adapting the sequence, pacing, and escalation of events in response to learner actions, with a focus on the learning objectives and assessment criteria in mind. By prioritizing learning-objective-driven episodes and technology constraints over strict timelines, the scenarios maintain pedagogical focus while remaining adaptable without compromising coherence or realism (Tena-Chollet et al., 2017). When designing operational episodes, instructors consider not only correct learner actions but also plausible suboptimal behaviors informed by their experience from live simulations, classroom teaching, and operational practice. Anticipated responses are documented together with suggested instructor

interventions, such as providing additional information, adjusting role-play interactions, modifying scenario tempo, or redirecting attention to critical cues. This preparation supports adaptive facilitation while maintaining the intended learning objectives embedded in each episode.

To support learners with limited experience with VS, Phase 3 also includes the development of a low-complexity familiarization scenario. This scenario should include relevant navigation and interaction to allow the learner to practice and gain confidence in a non-assessed and fail-safe setting, to support psychological safety before engaging with more complex training scenarios.

The main deliverables of Phase 3 are:

- Scenario briefs.
- Operator notes.
- Role-play instructions.
- Scenario software files.
- Technical and pedagogical test records.

Phase 3 deliverables establish the training content for the subsequent phases of OSM. The learning objectives and assessment criteria established in Phase 1 determine what the scenario(s) must elicit and make observable, while the organizational and technical conditions determined in Phase 2 define the practical boundaries for scenario complexity (e.g., extent of episodes), instructor involvement (e.g., number of roles), and the delivery format. Phase 3 therefore represents the integration of pedagogical requirements and operational constraints into executable simulation scenarios and provides the foundation for scenario preparation, briefing, execution, and assessment in the Phases 4–6.

4.5. OSM Phase 4 — train the instructor team

The purpose of Phase 4 is to prepare instructor teams to independently deliver VS training with technical proficiency, pedagogical coherence, coordinated teamwork, and the ability to facilitate learning, maintain learner presence, and conduct assessment and feedback activities (building on Phases 1–3). The phase consists of three complementary activities: (1) pedagogical preparation, (2) individual instructor preparation, and (3) instructor-team rehearsal.

Beginning with pedagogical preparation, a shared pedagogical understanding of simulation-based training is developed. Particular emphasis is placed on understanding how learning objectives, assessment criteria, scenario design, and instructor actions interact to shape learner experiences and support experiential learning (Kolb, 1984). Instructors familiarize themselves with the learning objectives and assessment criteria defined in Phase 1 and the scenarios and supporting materials developed in Phase 3. This preparation establishes a common pedagogical foundation and helps ensure that learning objectives remain central during subsequent training delivery and assessment activities.

Next, individual instructor preparation allow the instructors to gain practical experience with the technology and become familiar with the responsibilities associated with the instructional roles, including operator, role-player, and assessor. Instructors are encouraged to practice multiple roles through role rotation to develop a shared understanding of role interdependencies, scenario dynamics, and opportunities to observe and assess learner performance. Particular attention is given to the operator role, as it often represents a bottleneck for implementation and sustained use of VS. Ensuring that multiple instructors can perform the role reduces dependency on individual instructors, strengthens organizational resilience, and supports the long-term sustainability of VS use.

Once instructors are familiar with both the pedagogical foundations and their individual responsibilities, instructor-team rehearsals are conducted using complete training scenarios. These rehearsals focus on team coordination, communication procedures, scenario pacing, transitions between episodes, learner experience, learner experience, and the consistent application of assessment criteria. Consistent with CAPM (Polikarpus et al., 2021), rehearsals provide opportunities to identify challenges in scenario execution and refine instructional practices before learner participation, while considerations from the more general FIVE (Slater & Wilbur, 1997) support evaluating interaction and the sense of presence.

Through repeated rehearsals, instructors can gradually develop shared verbal and non-verbal communication practices to support synchronization during scenario delivery, e.g., standardized phrases and commonly understood gestures for communicating pacing, role-play injects, assessment-related observations, and emerging support needs. Such practices help maintain a coherent learner experience, contribute to a high sense of presence, and improved team performance. Rehearsals also provide an opportunity to verify that the training schedule established in Phase 2 allows sufficient time for scenario execution, briefing activities, and feedback while remaining compatible with organizational constraints. Rehearsals continue until the team demonstrates sufficient proficiency to conduct training independently. Representative learners are invited whenever possible to support realistic rehearsals and prepare instructors for likely learner behaviors and interactions. Particular attention is given to coordination between the operator and assessor roles, including the ability to adapt scenario progression in response to learner actions while maintaining alignment with the learning objectives and assessment criteria. In addition to the main training scenarios, instructors rehearse the learner pre-briefing and familiarization scenario, as well as conducting assessment and providing feedback based on the assessment criteria. These rehearsals prepare instructors to support learners in all aspects of the simulation, e.g., navigation, using communications systems, and understanding relevant system functions before entering training activities. Although instructors may be experienced in providing feedback in other training formats, they should also receive dedicated training in conducting feedback and assessment within VS-supported learning environments.

The main deliverables of Phase 4 are:

- Instructor training records and rehearsal logs.
- Refined operator-guide, role-play instructions, and supporting documents.
- Established communication, synchronization, and coordination procedures.
- Practiced evidence collection and feedback routines.
- An instructor team capable of independently delivering the planned training.

The Phase 4 deliverables transform the training design developed in Phase 3 into coordinated instructional practice. Shared pedagogical understanding, individual instructor competence, and team synchronization support that learning objectives remain central during training delivery. The proficiency and coordination achieved in Phase 4 are essential for the reliable execution of scenarios in Phase 5 and for

consistent, evidence-based performance assessment in subsequent phases.

4.6. OSM Phase 5 — conduct the VS training

The purpose of Phase 5 is to deliver VS training sessions in a controlled, pedagogically aligned manner that supports learner engagement, performance assessment, and psychological safety (Edmondson, 1999). This phase operationalizes the planning, preparation, and scenario development completed in the preceding phases, where instructional design, instructor competence, and technical infrastructure converge in real-time training practice. Prior to learner participation, the instructor team conducts technical and environmental readiness checks to verify that the training environment is fully operational. Using the established checklist, instructors confirms that workstations, displays, communications systems, software configurations, and scenario files are functioning as intended. These checks minimize the interruptions during introduction and training and help ensure a stable learning environment.

The pre-briefing is typically led by a designated instructor who is responsible for ensuring that learners have received and understood all relevant information before entering the simulation. This includes confirming that learning objectives, assessment procedures, communication structures, and technical aspects have been explained and that any remaining questions have been addressed. In addition to conveying information, the instructor have to actively work on reducing the learners' uncertainty and anxiety by establishing a supportive and welcoming atmosphere. Informal conversation about previous training experiences, operational roles, or professional background can help build trust and create psychological readiness contributing to the learners' comfort and psychological safety. The goal is to ensure that learners feel informed, supported, and ready to engage with the simulation before the training begins. For learners with limited prior experience of VS, an introductory, non-assessed scenario may be conducted to allow to practice navigation, communication, and basic system interactions.

When both the learner and the instructor team are ready, the training scenario is initiated. During the scenario execution, the instructor team coordinates the flow, pacing, and adaptations in response to learner actions, within the boundaries defined by the learning objectives, assessment criteria, and scheduled training time. The operator manages scenario dynamics and system functionalities, role-player portray relevant roles and provide contextual information in relation to the scenario and adjusted to the learners' actions. The assessor observes the learner performance and, when appropriate, may suggest adjustment for the role-play interactions or scenario events to ensure that specific objectives are adequately addressed. Throughout the session, the instructor team works collaboratively to promote a high sense of presence (Slater & Wilbur, 1997) and engagement by maintaining realistic role-playing, appropriate pacing, coherent information flow, and suitable escalation of scenario events while preserving the pedagogical intent of each operational episode. Continuous verbal and non-verbal communication within the instructor team supports coordination, shared situation awareness, and the management of unexpected events, such as unpredicted learner actions or technical issues, without disrupting the learner experience. This coordination enables instructors to maintain focus on targeted competences and observable behaviors while ensuring consistency between scenario delivery and assessment practice.

The main deliverables of Phase 5 are:

- Completed training sessions.
- Instructor observation notes.
- Learner performance evidence.
- Session logs and technical records.

The Phase 5 deliverables constitute the primary evidence generated through training delivery in real training settings and provide the basis for feedback, reflection, and performance assessment activities to be conducted in conducted in Phase 6.

4.7. OSM Phase 6 — conduct learner feedback, assessment, and evaluate instructor team performance

The purpose of Phase 6 is to conduct learner feedback and assessment based on performance evidence gathered during scenario execution and to support the continuous development of the instructor team. This phase links the learning objectives and assessment criteria established in Phase 1 to the evidence collected in Phase 5, ensuring that feedback aligns to assessment and are evidence-based, transparent (Bloom et al., 1956), supporting both learning and evaluation.

Following scenario completion, all instructors involved contribute their observations related to the learning objectives and assessment criteria. Because instructors observe learner performance from different roles and perspectives, throughout the scenario execution, they may observe different aspects of situational awareness, communication patterns, leadership behaviors, decision-making (Argyris, 1977) cues, and operational reasoning (e.g., Effective Command, 2025; Keskitalo & Ruokamo, 2015; Polikarpus et al., 2021). Prior to learner feedback session, these observations are discussed and consolidated within the instructor team, facilitated by the assessor. This collaborative evidence-gathering may provide a borader and more robust foundation for assessment than would be possible from a single perspective.

The OSM feedback begins with a structured learner reflection. Learners are encouraged to self-reflect on their experience and performance in relation to the learning objectives before receiving the assessors' feedback. This supports metacognitive development and encourages learners to critically examine the assumptions and reasoning underlying their actions, consistent with principles of experiential learning (Kolb, 2014) double-loop learning (Argyris, 1977). Instructors may also contribute with observations that extend beyond directly observable actions and outcomes. For example, a role-player may have exhibited hesitation, uncertainty, or reduced confidence during interactions with the learner, even though the learner ultimately made an appropriate decision. Such observations can provide valuable insights into the learner's reasoning and decision-making processes and should be incorporated into the feedback discussion. By integrating multiple instructor perspectives, the assessment can more accurately capture both what the learner did and how the learner approached the situation.

Feedback to is grounded in assessment criteria and evidence indicators established in Phase 1. This helps ensure that the discussion remains focused on learning objectives and demonstrated behaviors rather than general impressions or outcome-based judgments. Particular emphasis is placed on the learner's reasoning and decision-making process, in accordance with Lamb et al. (2020).

When multiple scenarios are conducted sequentially, feedback and assessment are often primarily formative and developmental. Feedback

from earlier scenarios can be used to identify strengths, highlight areas for improvement, and support progressive learning across repeated training activities, consistent with experiential and applied learning principles (Bloom et al., 1956; Kolb, 1984).

In addition, the instructor team conducts an after-action review of its own performance during scenario delivery. This review examines team coordination, role-play quality, pacing, technical handling, communication patterns, and efficiency of evidence collection. The objective is developmental rather than evaluative, identifying opportunities for immediate improvements during ongoing training series, as well as longer-term improvements for future training cycles. Identified lessons are documented in a structured format to support continuity across training sessions and to provide inputs to the systematic improvement activities of Phase 7. In this way, Phase 6 contributes not only to learners' development but also to maintaining the reliability, consistency, and pedagogical integrity of the entire OSM cycle.

The main deliverables of this Phase 6 are:

· Learner assessment and feedback documentation.
· Consolidated performance evidence.
· Instructor after-action review notes.
· Improvement log and recommendations.

The deliverables of Phase 6 conclude the learner-focused assessment cycle and initiate the improvement cycle for future training activities. Learner assessment documentation supports learner reflection and competence development, while instructor after-action review may contribute to the next training session and continuous improvement activities for Phase 7.

4.8. OSM Phase 7 — systematic improvement

The purpose of Phase 7 is to support the continuous improvement of the training practices using VS to maintain organizational capability required for its long-term use. This phase translates experiences and lessons learned into concrete improvements to training content, instructional practice, and organizational arrangements, while simultaneously preparing the next training.

Building on the feedback activities and instructor reflections conducted in Phase 6, Phase 7 ensures that lessons are reviewed and translated into documented updates. Scenario content is revised and version-controlled to maintain alignment with the learning objectives and assessment criteria, while also accommodating newly observed learner behaviors and emerging training needs. Scenario review and rotation schedules may be established to reduce instructor fatigue and maintain pedagogical engagement across repeated deliveries (Alenljung et al., 2026). Similarly, the familiarization scenarios and learner pre-briefing materials are updated to reflect changes in technology, communication procedures, or learner needs. At the organizational level, experiences from training delivery are reviewed in relation to staffing, schedules, instructor competence, and technical infrastructure. Adjustments may include allocating additional preparation time for instructors, scheduling regular rehearsal sessions to maintain instructor and operator proficiency, following an extended period without training (Alenljung et al., 2026) and qualifying multiple instructors for critical roles to ensure redundancy. Such measures reduce the risks associated with skill decay, dependence on individual instructors, and variability in training quality, thereby supporting the sustainable integration of VS within the organization's training system. Supporting documentation and training resources are also maintained and updated. This includes scenario packages, assessment templates, communication protocols, learner pre-briefing materials, technical checklists, and onboarding resources for new instructors. Systematic documentation of revisions provides traceability, promotes consistency across training cycles, and enables the gradual maturation of VS as a training practice within the organization.

The main deliverables of Phase 7 are:

- Version-controlled scenario and assessment packages.
- Updated instructor schedules and rehearsal plans.
- Onboarding materials for new instructors.
- Updated training documentation and change log.
- Documented lessons learned and organizational improvement actions.

These Phase 7 deliverables establish the structural and pedagogical foundations for subsequent training with OSM.

5. Reflections from the iterations and synthesis

This section synthesizes empirical evidence from the eight AR iterations and examines how OSM functioned and evolved in different real organizational contexts. Building on the observations summarized in Appendix B, this section first describes the motivation for successive iterations and then analyzes recurring patterns in the introduction and use of VS. The focus is on how OSM phases functioned over time and how repeated use contributed to increasing organizational ownership and sustainable use of VS.

5.1. Motivations and learning from practice to choose the iterations

The iterations took place in organizations with different resource conditions, levels of VS experience, and training objectives across IC1–IC3 training. Overall, the findings indicate a progression from initial exploration of VS use towards a more robust and transferable method for integrating VS into organizational training practice. OSM was applied in contexts ranging from IC1 training and examinations to more complex IC2 and IC3 training activities. Table 2 summarizes the primary motivations for each iteration. While organizational context, objectives, and preconditions differed, all iterations contributed to the refinement of OSM by revealing practical requirements, recurring challenges, and opportunities for improvement. Detailed phase-based observations are presented in Appendix B.

Table 2
Primary motivations for choosing iterations.

I1	Initial development under time pressure revealed the need for clear applied learning objectives aligned to scenario design, technical readiness, and instructor coordination.
I2	Establishing foundations: applied objectives, basic technology setup, structured instructor training, pre-briefing, and assessment practices.
I3	Scaling and standardization: parallel instructor teams, structured scenario documentation, and improved coordination practices.
I4	Adapting to multi-agency scenarios; refining role distribution and instructor coordination.
I5	Adapting method to limited resources; simplified setup and assessment, and demonstrating robustness.
I6	Transfer to a new organization; validating feasibility with limited preparation and structured support.
I7	Increased instructor fluency, autonomy, and reduced reliance on researcher support.
I8	Stabilization and optimization; high coordination, efficient execution, and organizational integration.

5.2. Cross-iteration patterns and refinement

Application of OSM across new organizations, instructor teams, and training contexts revealed both recurrent strengths and recurrent challenges. Structured scenario documentation improved consistency across instructor teams (I3 onward), while clearer role distribution, particularly assessor-led pacing and shared focus on learning objectives during training delivery, strengthened alignment between scenario execution and assessment. At the same time, recurring bottlenecks highlighted important organizational prerequisites for sustainable VS use, including

technical competence associated with the operator role and the need to match and strengthen assessor competence to the command level being trained. Analysis of the eight iterations shows a clear progression in how OSM was understood, applied, and stabilized. During the first iteration (I1), the phases primarily served to establish prerequisites for VS integration. Learning objectives and assessment criteria required clarification (Phase 1), technical requirements and resource constraints were identified (Phase 2), instructor coordination proved challenging (Phase 4), and assessment practices lacked structure (Phase 1 and 6). Instructor reflections highlighted the need for clearer planning, practical rehearsal, and better synchronization among instructional roles: "*We were all doing our part, but it took some practice together to become and keep synchronized… Viewing the teamwork as an orchestra made us work more together, focusing on the learner… while keeping the flow realistic*."

During the middle iterations (I2–I4), the phases supported improved coordination and adaptation. Learning objectives guided scenario design more directly, organizational constraints were incorporated into planning and scenario design, and structured documentation improved consistency across teams. Instructor training evolved from learning individual roles towards developing coordinated team performance through rehearsal, role rotation, and use of the orchestra metaphor. Assessment practices also became more structured through templates, clearer criteria, and contributions from multiple instructors.

During later iterations (I5–I8), the phases focused less on establishing functionality and more on refinement, contextual adaptation, and continuous improvement. Learning objectives required minimal adjustments, resource requirements became better understood, the instructor team established effective coordination more rapidly, and assessment practices became increasingly efficient and trusted. Organizations also required less external support and demonstrated increasing ability to adapt and refine training activities independently.

Across all iterations, several themes recurred: the importance of aligning learning objectives, scenario design, and assessment; the central role of instructor coordination and rehearsal; and the influence of organizational prerequisites such as resources, staffing, scheduling, and technical readiness. Together, these themes acknowledge that successful introduction and use of VS depend not only on technological capabilities but also on the coordinated development of instructional and organizational practices (Bull & Bell, 2009; Fomin et al., 2024; Kolb, 2014; Makransky & Petersen, 2021).

5.3. Aligning learning goals to VS

A central mechanism of OSM was the alignment between learning objectives, instructor roles, scenario design, assessment, and organizational conditions. Early iterations showed that unclear, overly broad, or low-level learning objectives created uncertainty regarding both the role of VS in IC training and how scenarios should be designed. It became difficult to determine the appropriate scenario complexity and scope, required interactions, and involved roles. Strengthening Phase 1 through applied and observable learning objectives provided clearer guidance for scenario design (Phase 3) and clarified how VS could complement LS. As one instructor noted, VS requires stronger alignment focus on learning objectives and their alignment with scenario design, because VS flexibility can otherwise lead to unfocused scenarios, since "*In LS you cannot simulate fully dynamic scenarios… In VS you can potentially create any scenario… With this opportunity, you need to start with the learning objectives… not become tempted to do all kinds of 'fun' scenarios.*" Alignment between organizational preconditions and scenario design was equally important. Resource constraints, instructor availability, class size, scheduling, and technical infrastructure directly affected the complexity, scope, and delivery format of the scenarios. These conditions determined how many roles could be actively supported, the number and duration of operational episodes, and the level of dynamic interactions that the instructor team could sustain. OSM made these dependencies explicit, supporting instructors' understanding of how scenarios can be designed both educationally purposeful and operationally feasible, as verbalized by one instructor: "*Just because we can make a scenario more complex does not mean we should. The learning objectives, instructor capacity, and available time need to be considered when deciding the level of challenge. Otherwise, there is a risk that learners become overwhelmed and that important opportunities are lost.*" Nevertheless, instructors' effort consistently needed to remain focused on learning objectives and the quality of training during coordinated activities across OSM phases. Early difficulties with synchronization, communication, and role clarity underscored the need for both individual technical preparation and team-level rehearsal (Phase 4).

Over time, shared coordination strategies and role rotation supported more fluent teamwork and allowed realistic scenario progression, coherent information flow, and timely responses to learner actions. The assessor role and the quality of assessment (Phase 6) were similarly dependent on effective team coordination. Maintaining awareness of scenario progression, role-play interactions, and learner actions enabled assessors to identify evidence of learning objectives and assessment criteria. In some situations, assessors influenced pacing, information flow, or role-play interactions to ensure that key competencies could be observed and meaningfully evaluated. As one instructor reflected, "*if we sense that a learning objective has not really been triggered yet, we need to coordinate and adjust the role-play or provide additional cues so that the learner gets an opportunity to demonstrate the competence we are supposed to assess*." This observation shows how the orchestration metaphor helped the instructors to conceptualize the interdependence of their roles and collective performance in supporting the learners' presence and engagement. As one instructor explained, training success depended on "*good timing, flow, flexibility, and real-time adaptation with the focus on learning objectives*," while another noted that "*if one of us misses what is happening, everything slows down immediately*." Taken together, these observations illustrate how the quality of scenario delivery emerged from coordinated team performance rather than individual instructor actions. Several instructors also noted that adding a new instructor to the team often required a period of rehearsal and adaptation before the team regained its previous level of synchronization, coordination, and performance. This illustrates the importance of aligning instructor preparation and rehearsal time (Phase 2) with instructor team development and practice (Phase 4).

The examples above, and from Appendix B, illustrate instructors' reflections and behavior, indicating that, in the context of OSM-supported VS adaptation to varying organizational conditions, VS training can be performed and aligned to training objectives, despite differences in resources, instructor experiences, and training objectives. The phases were beneficial for adjusting scenario complexity, pacing, and delivery format while maintaining alignment with both learning objectives and organizational constraints.

5.4. Organizational ownership for sustainable use of VS

Across iterations, instructor teams moved from reliance on external support towards increasing organizational ownership of VS. Early cycles required assistance with technical setup and coordination, whereas later cycles demonstrated increased confidence and competence in planning, delivering, assessing, and refining training activities. This progression was supported particularly by the systematic improvement activities of (Phase 7).

Evidence suggests that organizational ownership emerged gradually over time. Repeated use of OSM was associated with changes in how individual phases were enacted over time, based on experiences from earlier iterations. As shown in Appendix B, the early iteration (I1) involved experimentation, providing the basis for adjustments related to technical preparation, instructor coordination, training schedules, and the alignment of learning objectives with scenario design. Instructor reflections highlighted the need for clearer planning and greater coordination among trainer roles.

Progression was visible in Phase 4, where instructor training involved learning the technology and clarifying responsibilities (I1), improved synchronization through rehearsal, role rotation, and use of the orchestra metaphor (I2–I4), and finally highly coordinated and autonomous team performance requiring minimal preparation (I5–I8). The VS experts observed improved team synchronization during the second iteration in each organization. Instructors increasingly used terminology from the orchestra metaphor in OSM to discuss coordination and continuous improvement within the instructor team. Using verbal and non-verbal signals agreed upon in advance, the assessor indicated when sufficient evidence had been collected or when additional assessment opportunities were needed. This prompted the role-players or operator to adjust their interactions, introduce new cues, or modify the scenario pace to support both learning and assessment without disrupting the learner experience.

A similar development was observed in Phase 6. Early iterations revealed concerns about assessment quality, assessor credibility, and inconsistent evidence collection in relation to learning objectives. In response, assessment templates were developed and progressively aligned with learning objectives and scenario episodes. Multiple instructors' perspectives increasingly contributed to evidence gathering, enabling assessments to be based on broader and more reliable observations. During the later iterations, these practices resulted in more efficient documentation, more consistent feedback, and greater confidence among instructors in the assessment process. The VS experts also observed an increasing understanding of the distinctive opportunities and constraints of VS in relation to traditional LS. According to the experts, OSM helped shift the focus from what could be simulated to what should be simulated to most efficiently support the intended learning outcomes.

Repeated use of OSM therefore strengthened coordination between phases and increased the organizations' ability to independently plan, deliver, assess, and refine training activities in VS. Rather than focusing on technical implementation, later iterations increasingly addressed refinement, adaptation, and continuous improvement of practice. Evidence of increasing organizational ownership was also visible in the organizations' future intentions and independent use of VS. MCF and Skaraborg planned continued use of VS within regular training structures and independently conducted training in VS after the research period, whereas Nerike and Karlstad primarily focused on exploration and evaluating the potential of VS, consistent with their original objectives. Although organizations progressed at different rates, all

Table 3
Frameworks and methods motivating the definition of OSM.

Framework/Model	Primary focus on/Usage context	Typical use in VS training	Relation to OSM Phases	Comparison to OSM
TAM (Davis and Venkatesh, 1995)	Technology acceptance (perceived usefulness and ease of use of new technologies in organizations). Used in various contexts.	Explains the acceptance and adoption of VS technologies by instructors and organizations.	Phase 2	Focuses on end-user adoption rather than how VS training is implemented and managed in practice.
Kirkpatrick model (Kirkpatrick & Kirkpatrick, 2009)	Evaluation of training outcomes (reaction, learning, behavior, results). Used mainly in the health domain.	Used for assessing training effectiveness and impact (transfer of the learners to working contexts).	Phase 6, 7	Focus on learners and evaluation rather than how training is designed, implemented, and managed.
TPACK (Bull & Bell, 2009)	Integration of technological, pedagogical, and content knowledge. Developed for educational settings and vocational training.	Supports instructors' understanding of integrating pedagogy and technology into training design (Polikarpus et al., 2023).	Phase 1, 4	Does not provide operational guidance for planning, coordination, delivery, or iterative improvement of training
SCM (Lui & Goel, 2022)	The situated cognition model connects VR technology used in a context to situated learning. Developed for understanding hospitality.	Support understanding the role of situational awareness for designing training scenarios. Examines the role of realistic, context-rich training.	Phase 3	It is mainly from the learners' perspective, focusing on proper support for learning rather than the multifaceted duties of instructors.
TRIFECTA (De Vaujany et al., 2018)	Understanding the three core elements (rules, practices, and information technology) requires seamless (and efficient) use and interpretation of technologies.	Explains the challenges in implementing complex technologies such as the VS role in organizational contexts (Fomin et al., 2024).	Phase 2, 4, 7	Provides a conceptual understanding of the complexity of managing VS-based education and the aspects organizations need to handle (not educators), but does not offer structured implementation guidance.
CAMIL (Makransky & Petersen, 2021)	Examines cognitive and affective processes in immersive learning across different domains, primarily in education and science.	Explains how presence, engagement, and cognitive load influence learning in VS.	Phase 5, 6	Focuses on learner experience rather than instructor-led implementation and organizational integration, and, as of today, is not used for emergency preparedness training.
FIVE (Slater & Wilbur, 1997)	Defines immersion through inclusiveness, extensiveness, surroundingness, vividness, body matching, and plot autonomy. Not domain-specific.	Understand the presence of multiple users in shared graphical environments, in general. Influencing, considering the role of social aspects for VR collaboration (Heldal, 2007).	Phase 3,5,6	FIVE helps to understand different aspects of the VS used. For example, how fidelity ("realism" of the object, environment, and interactions) influences presence, situational awareness, and performance in single- and multiuser settings.
SBLE (Keskitalo & Ruokamo, 2015)	Support pedagogical design for simulation-based learning environments. It is developed for healthcare education.	Emphasizes three core components: orientation, simulation, and debriefing for learner preparation, guided facilitation, and reflection.	Phase 3, 5	Does not address instructor coordination, organizational constraints, or the full training lifecycle.
CAPM (Polikarpus et al., 2021)	Supports instructors in collaborative scenario authoring. Defined for firefighter IC training.	Supports co-design of simulation scenarios and learning activities. It is used in Estonia.	Phase 3	Does not address implementation, execution, assessment, or continuous improvement of training.
Effective Command (Lamb et al., 2020) (IC competence framework)	Development of operational decision-making, leadership, and coordination skills. It is defined for practical settings, mainly for firefighter training.	Defines learning objectives and performance expectations for incident commanders. It is applied in several environments (Polikarpus et al., 2019; Reis & Neves, 2019).	Phase 1, 6	Defines *what* should be trained, but not how VS training should be implemented or coordinated by instructors.
OSM	Structured implementation and management of VS training.	Supports the full lifecycle: planning, design, coordination, delivery, assessment, and continuous improvement.	Phases 1 – 7	Integrates multiple aspects but depends on organizational readiness and instructor competence to conduct training. Allows training transfer between the different VS-supported training.

demonstrated increasing local competence and responsibility for planning, delivering, assessing, and improving VS-based training.

Across the examined organizations, organizational ownership emerged through the gradual strengthening of cross-phase alignment, shared instructor routines, and systematic learning across iterations. OSM therefore functioned not only as a method for delivering VS training, but also as a mechanism for developing instructor competence and organizational capability needed for its sustainable use.

6. Discussion, limitations, and future work

This section discusses OSM in relation to existing frameworks and methods, presented in Section 2.3, reflects on limitations associated with development and evaluation, and identifies directions for future research.

6.1. OSM in relation to other frameworks and methods supporting VS

To position OSM in relation to existing frameworks and methodologies relevant to the introduction and use of VS, Table 3 contrasts their primary focus, typical use, in relation to OSM. The overarching aim is to more clearly delineate their respective roles and to justify the subsequent definition of OSM.

The comparison in Table 3 demonstrates that existing frameworks primarily address discrete aspects of VS, such as technology acceptance and adoption, pedagogical considerations in scenario design, learner processes, and assessment. OSM contrast to the other frameworks, provides structured support for introducing, delivering, assessing, and continuously improving VS across different training objectives, command levels, and organizational contexts. Successful implementation depends on organizational readiness, instructor competence, and available resources. OSM conceptualizes VS as an orchestrated training practice and provides structured support for the introduction and management of VS. An important framework that motivates several aspects of OSM, including the benefits of viewing VS as a weakly structured system (one that necessitates organizational rule and policy changes for efficient use), is the Trifecta concept (De Vaujany et al., 2018). Trifecta highlights the inherent complexity of integrating technological solutions within organizations responsible for education and training. Similar to Trifecta, OSM illustrates that the successful introduction of VS is shaped less by the technology itself than by the organizational and instructional systems into which it is embedded. Trifecta provides a conceptual understanding of organizational complexity, which OSM translates into a practical method for instructors and organizations responsible for delivering IC training using VS.

6.2. Limitations and future work

Several limitations should be considered when interpreting the findings. The AR approach involved the researchers' dual role as practitioners in close collaboration with practitioners throughout the development and refinement of the method. While this strengthened the practical relevance of the method and enabled iterative refinement based on real-world use, it introduced a potential risk of researcher influence on both the intervention and interpretation of findings. This risk was mitigated through repeated iterations across four organizations, continuous reflection with participating instructors, and documentation of both challenges and successful outcomes, and considering reflections from research. The analysis primarily relies on qualitative data derived from instructor observations, reflections, and iterative development activities. While this aligns with the instructor-focused purpose, the absence of independent measures of trainee learning outcomes limits the ability to assess training effectiveness and compare OSM with alternative approaches. Another limitation concerns the distinction between OSM's core and adaptable components. Although the seven phases remain stable across the study iterations, their implementation varied depending on the organizational size, available resources, training objectives, instructor competence, and local constraints. Consequently, some activities, deliverables, and supporting checklist items may require adaptation when OSM is transferred to new settings. For example, several phases may be reused with limited modification when training similar groups, whereas application to other emergency professions may require different assessment practices, instructional roles, or scenario structures. Furthermore, all iterations were conducted within Swedish FRS organizations and used the same technology (XVR).

Consequently, the applicability of OSM to other emergency management domains, organizational settings, technological environments, and national training systems remains to be investigated. Future applications may therefore motivate merging, subdividing, or redefining individual phases while preserving the underlying OSM principles.

Future research should examine the applicability of OSM across different technologies, emergency management domains, organizational settings, and national training systems. Such studies would help assess the transferability of OSM and clarify which components are essential and which can be adapted to local needs. This would identify the distinction between OSM's core and adaptable components, enabling greater consistency while preserving flexibility across different contexts. In addition, future studies should investigate trainee learning outcomes and compare VS training supported by OSM with alternative approaches for introduction and use of VS. This would complement the instructor-focused perspective and provide further knowledge regarding the educational value of the method, but also investigate the resources required and the cost-benefit implications of introducing and sustaining VS using OSM.

Finally, longitudinal studies how OSM would influence instructor competence, organizational learning, and the sustained integration of VS over extended periods of use. Such research could also explore the resources, costs, and organizational conditions required to maintain and further develop VS training practices.

7. Conclusions

This paper presents the OSM, a structured and practice-based method developed to support FRS organizations in introducing, delivering, and continuously improving VS for IC training. Compared to approaches that primarily focus on technology adoption, learner experience and outcomes, or instructional design, OSM addresses a complementary challange: developing instructor and organizational capability for sustainable integration of VS into training practice.

Developed through eight AR iterations across four Swedish organizations, OSM conceptualizes the introduction and use of VS as an orchestrated process in which learning objectives, instructor roles, scenario design, training delivery, feedback and assessment activities, and continuous improvement are aligned with organizational conditions and

available resources. Examining the use of OSM across several IC training contexts indicates that such alignment is critical for introducing and managing VS, particularly in organizations with limited prior experience of VR technologies and VS training format. The findings illustrates how OSM can support instructors in systematically planning, delivering, evaluating, and refining VS training while enabling organizations to develop ownership of both training process and the technology.

The main contributions are arguments for the development of OSM, the method itself, and a set of practical tools, phase workflows, and checklists that support its application in practice. Additionally, the paper provides an improved understanding of OSM in contrast to current methods and frameworks in introduction and using VS in organizations.

Although developed and evaluated within Swedish IC training, the principles underlying OSM may be relevant to a broader range of simulation-based training contexts. While OSM complements existing frameworks concerned with technology adoption, instructional design, or evaluation of VS training format, future research should examine its applicability across different organizations and technological settings, further distinguish between core and adaptable components of the method, and investigate its long-term effects on instructor competence, organizational ownership, and the sustainable integration of VS. Taken together, the results highlight that successful use of VS depends not only on the technology itself, but also on the organizational and instructional capabilities required to continuously adapt, maintain, and improve it over time.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

All written content in this thesis was drafted by us, ensuring that ideas, reflections, and conclusions are our own. AI tools (Grammarly, ChatGPT, Copilot), were used afterward to correct grammar, improve sentence structure, and enhance the overall flow and readability of the text.

CRediT authorship contribution statement

Cecilia Hammar Wijkmark: Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ilona Heldal:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Sune Fankvist:** Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Operational details for the eight iterations

Column 1: Iteration number (I#).
Column 2: VS set up location, dedicated (D) or temporary (T) facilities.
Column 3: IC training level (IC1, IC2, IC3), Examination (E), Annual development (A), Basic training (B).
Column 4: Number of instructors (number of instructors with previous VS experiences from previous iteration or another context.
Column 5; Technical set up, number of computers with simulation software license.
Column 6: Number of learners planned to be trained by the instructor team during the current year/scheduled training time per learner.
Column 7: Number of scenarios (the duration per scenario/ description of complexity).
Column 8: Duration of instructor team training based on OSM.

I#	D/T	Lev.	# instr. (VS exp.)	# computers	# learners/training time	# Scenarios (duration/complexity)	Instructor training duration
I1	D	IC1E	5 (2)	5	60/200 min.	4 (avg. 45 min./complete IC1 incidents)	1 week
I2	D	IC1A	8 (0)	3	85/180 min.	2 (60 min./complete IC1 incidents)	1 week
I3	D	IC1B	8 (2)	6	180/30 min.	6 (15–30 min./IC1 initial response)	3 days
I4	T	IC2, IC3A	10 (8)	3	25/120 min.	2 (60 min./complex multi-agency)	2 days
I5	T	IC2, IC3A	5 (2)	5	18/120 min.	3 (20–40 min./IC2/IC3 incidents)	1 day
I6	T	IC1, IC2, IC3A	6 (2)	5	16/120 min.	3 (20–40 min./advanced IC1, IC2, IC3 incidents)	1 day
I7	T	IC1A	5 (5)	5	18/120 min	3 (20–40 min./complete IC1 Incidents)	1 day
I8	T	IC1,IC2,IC3A	6 (4)	5	16/120 min	2 (60 min./complex multi-agency incidents)	1 day

Appendix B. Evidence structured by the OSM phases

The table summarizes evidence across all OSM phases (Columns 2–8) and iterations (Rows I1-I8). The warning symbol (⚠) denotes challenges or issues requiring attention followed by the planned improvement (–>), and a check mark (✓) indicates evidence of effective functioning. The table was used to identify patterns of variation, improvement and stabilization across iterations. The table was used to identify patterns of variation, adaptation, improvement, and increasing organizational ownership across early (I1–I2), middle (I3–I5), and later (I6–I8) iterations.

I#	Phase 1. Define learning objectives and assessment criteria	Phase 2. Define resources and preconditions	Phase 3. Design scenarios and build them in VS	Phase 4. Train the instructor team	Phase 5. Conduct VS training	Phase 6. Conduct learner feedback, assessment and evaluate instructor team performance	Phase 7. Systematic improvement	Instructor comments	Key iteration reflections
I1	⚠ Objectives too descriptive → Reformulated to applied level; explicit assessment criteria needed.⚠ Assessment documentation missing → Structured assessment documentation developed.	⚠ Technical setup and connectivity issues → Structured resource planning, technical checklist, and identification of required technical competence introduced.	✓ Scenarios aligned with assessment criteria for examination.	⚠ Instructor training insufficient for technology familiarization and team coordination. ⚠ Using instructors as learners during rehearsal proved insufficient.→ Structured rehearsals with representative learners and shared verbal and non-verbal communication routines (e.g., hand signals) introduced.	✓ Introduction video call was introduced to reduce uncertainties and anxiety and improve psychological safety in the examination situation.	✓ Structured debriefs and all instructor team input to assessment improved assessment quality and time efficiency.	✓ Early lessons informed later improvements in delivery synchronization, technical preparation, assessment, and learner pre-briefing.	"Requires synchronized teamwork like an orchestra", "It is important that do my best to reduce the students' stress by explain how this works and also ensure them that this is a safe environment and we all have focus on their learning"	Ensure sufficient instructor training time with representative learners to strengthen technical and coordination flow.
I2	✓ Applied learning objectives (Bloom ≥ 3) improved alignment.	✓ Minimum technical setup defined based on constraints.	⚠ Limited technical resources → scenario complexity, role distribution, and content adapted to local constraints.	⚠ Coordination challenges → dedicated training week introduced to improve synchronization. ⚠ Operator role created bottlenecks → cross-training of instructor roles encouraged.	⚠ Training schedule too intensive → instructor workload and delivery schedule revised.	✓ Increased assessor confidence through structured assessment criteria and instructor team reflection contribution to assessor,	✓ Continuous scenario refinement based on learner actions and assessment outcomes.	"Clear objectives and criteria help learning without harsh critique", "after running five scenarios I am tired and not sure about what I have said or not said to the student)	Improve the use of the orchestra metaphor in instructor training. Adjust training schedule and improve planning of instructor workload.
I3	✓ Improved alignment with new national IC programme learning objectives.	✓ Technical setup and schedule adapted for parallel instructor teams. ⚠ Limited scheduled training time constrained scenario scope. ⚠ New inexperienced instructors required onboarding and additional rehearsal.	✓ Structured documentation ensured consistency across teams.	✓ Orchestra metaphor, pedagogical grounding, representative learners, and increased rehearsal time improved team synchronization and performance.	✓ Clarification of assessor role improved scenario pacing and team coordination.	✓ Assessment quality improved through templates and clearer criteria.	✓ Cross-team reflection supported improvements across multiple OSM phases.	"Repeated scenarios allow fine-tuning and experimentation", "with the limited scenario time, we need to be very focused on the learning objectives and have not time to include anything extra"	Improve onboarding for new instructors, strengthen documentation and templates, scenario design is dependent on scheduled training time.
I4	✓ Applied learning objectives aligned with VS.	✓ Resource planning adapted to recurring constraints and onboarding of new	✓ Scenarios scaled to complex multi-agency events.	✓ Experienced teams required less training.	✓ Improved assessor–operator balance. Assessor increasingly guided	⚠ Assessor credibility varied across command levels → assessment	✓ Iterative refinement stabilized scenarios, role-play	"Structured assessment help strengthen my credibility when assessing higher command levels"	Ensure appropriate allocation of assessor and operator roles.

(*continued on next page*)

(*continued*)

I#	Phase 1. Define learning objectives and assessment criteria	Phase 2. Define resources and preconditions	Phase 3. Design scenarios and build them in VS	Phase 4. Train the instructor team	Phase 5. Conduct VS training	Phase 6. Conduct learner feedback, assessment and evaluate instructor team performance	Phase 7. Systematic improvement	Instructor comments	Key iteration reflections
		instructors, particularly for the operator role.			scenario pacing and assessment opportunities.	documentation and evidence structures strengthened.	activities, and team synchronization.		
I5	✓ New applied objectives developed (no prior objectives available).	✓ Simplified setup adapted to constraints and available instructor team size and previous experience.	✓ Scenarios based on operational experiences aligned with learning objectives and local training needs.	⚠ Novice instructors and limited rehearsal time → experienced VS instructors initially embedded within the instructor team.	✓ Role-play guidance based on instructors' operational experience contributed to learner presence and realism.	✓ Simplified assessment templates improved feasibility.	✓ Demonstrated that OSM remained workable under constrained organizational conditions.	"This method enabled us a quick start to use VS"	Ensure instructor rehearsal involving learners with relevant competence.
I6	✓ Objectives adapted from previous iteration.	✓ Structured learner pre-briefings improved readiness and minimized start up time under constraints.	✓ Scenarios successfully adapted to local context.	✓ One-day instructor training shown feasible when supported by preparation materials and experienced instructors.	✓ Improved learner psychological safety by pre-briefing was acknowledged. Increased instructor independence, reduced reliance on researcher support.	✓ Strengthened assessment confidence and structure.	✓ Improved understanding of how systematic updates and improvements could be coordinated across OSM phases.	"Structured method supports continuous improvement"	One-day introduction viable with preparation and experienced roles.
I7	✓ Applied objectives developed (no prior objectives).	✓ Sufficient minimal technical and instructor resources.	✓ Scenario injects and pacing refined.	✓ creased fluency and coordination through repeated use and rehearsal.	✓ Autonomous and synchronized instructor team performance.	✓ Structured debriefs increased confidence.	✓ Reduced researcher dependence and increased organizational ownership of VS and improvement activities.	"Orchestra metaphor clarified coordination and complexity"	OSM functioning successfully across all phases.

Appendix C. OSM application checklist

Phase 1. Define learning objectives and assessment criteria

Metaphor: *Planning the concert by selecting the repertoire and specifying what constitutes an adequate quality of performance.*

- ▢ Learning objectives are observable, actionable, and aligned with curriculum or training requirements.
- ▢ Measurable verbs (e.g., Apply, Decide, Assess) used.
- ▢ Learning objective has at least one evidence indicator.
- ▢ Assessment criteria exist for each learning objective.
- ▢ Alignment check between learning objectives, evidence indicators, and assessment criteria.
- ▢ Learning objective-based event map drafted for scenario design.
- ▢ Rubrics define performance levels (pass/needs improvement).

Phase 2. Define resources and preconditions

Metaphor: *Setting the stage, ensuring that instruments, musicians, rehearsal time, and acoustics are prepared before the performance.*

- ▢ Required instructor roles covered (operator, assessor, role-player(s)).
- ▢ Protected time allocated for instructor training including rehearsals.
- ▢ Learner group size, composition, and training format defined.
- ▢ Available time for pre-briefing, familiarization, training, assessment, and feedback documented.
- ▢ Instructor staffing matched to learner numbers and assessment requirements.
- ▢ Instructor workload considerations.
- ▢ Training venue decided.
- ▢ Technical resources identified (hardware, software, communication systems, network access).
- ▢ Communication procedures/channels identified.
- ▢ Assessment and documentation requirements identified.
- ▢ Preconditions and constraints communicated to personnel involved in scenario design and instructor training.

Phase 3. Design scenarios and build them in VS

Metaphor: *Composing the score, structuring themes and solos while leaving room for controlled improvisation.*

- ▢ Each learning objective and assessment criteria mapped to at least one observable scenario event.
- ▢ Match scenario scope and complexity to learning objectives, instructor capacity, and available time.
- ▢ Scenario brief completed and documented (roles, timeline, injects, triggers).
- ▢ Operator guidance notes and role-play guidance prepared.
- ▢ Scenario technical complexity matched to available instructor capacity and training time.
- ▢ Scenario test-run completed (technical and pedagogical).
- ▢ Trainee pre-brief package prepared (learning objectives, technology introduction, expectations).

Phase 4. Train the instructor team

Metaphor: *Rehearse individually and as an ensemble, learn the parts, practice cues, and refine timing before the performance.*

- ▢ Shared pedagogical understanding established regarding how learning objectives, assessment criteria, scenario design, and instructor actions support learner development.
- ▢ Instructors briefed on learning objectives, evidence indicators, and assessment criteria.
- ▢ Scenario materials reviewed (scenario brief, operator notes, role-play scripts).
- ▢ Full scenario rehearsal completed (timing, pacing, coordination).
- ▢ Instructor roles practiced through rotation (operator, role-player, assessor).
- ▢ Shared communication and synchronization practiced established (verbal cues, hand signals, coordination routines).
- ▢ Instructor team demonstrates a shared understanding of how individual roles contribute to learning objectives and assessment.
- ▢ Scenario can be delivered within the scheduled training time.
- ▢ Feedback delivery practiced using the agreed assessment method.
- ▢ Operator proficiency demonstrated (or progressing toward automaticity).
- ▢ Redundancy established in the operator role (more than one instructor able to operate).
- ▢ Familiarization scenario rehearsed and aligned with pre-briefing.
- ▢ Rehearsal includes representative learners or realistic learner behavior when possible.
- ▢ Instructor team has established and tested a readiness checklist to verify technical systems (hardware, software, audio, communications, bandwidth).
- ▢ Technical support needs identified.

Phase 5. Conduct VS training

Metaphor: *Perform the concert; adjust tempo and dynamics to the learner's movements while keeping the score intact.*

- ▢ Training room prepared and free from distractions.
- ▢ Technical checklist completed before learner entry.
- ▢ Learner pre-briefing completed and understood.

☐ Learner concerns, uncertainties, and signs of anxiety have been addressed before training activities begin.☐ Familiarization scenario used when needed.
☐ Scenario pacing aligned with learning objectives and available time.
☐ Scenario delivered with engaged, timely, and synchronized role-play.
☐ Instructor team coordination remains stable throughout delivery.
☐ Performance evidence documented during training.
☐ Instructor interventions remain aligned with the learning objectives and assessment needs.

Phase 6. Perform learner feedback, assessment and evaluate instructor team performance

Metaphor: *Review the concert; identify strengths and areas for improvement, and refine future performances for both learners and the enable.*
☐ Assessment evidence collected and documented throughout training.
☐ Multiple instructor perspective contribute to assessment when relevant.
☐ Feedback conducted linked to the learning objectives and assessment criteria.
☐ Structured learner feedback completed and documented.
☐ Assessment decisions supported by documented observations and feedback dialogue with learner.
☐ Instructor team reflection on own performance conducted (coordination, role-play, assessment, and pacing reviewed).
☐ Improvement actions documented for training session and systematic improvements.
☐ review of future learning objectives, scenarios, and assessment needs.

Phase 7. Systematic improvement

Metaphor: *Refine and the score, ensuring that future performances benefit from accumulated experience and that the orchestra can perform reliably over time.*
☐ Lessons learned reviewed across all OSM phases.
☐ Scenario and assessment updates documented and versioned.
☐ Learning objective, scenario, and assessment alignment re-evaluated.
☐ Scenario rotation, update, and new development plan implemented.
☐ New instructor onboarding, training, team rehearsal and skill maintenance activities scheduled.
☐ Redundancy established for all roles.
☐ Updated materials archived and communicated.
☐ Preparation initiated for next training cycle.

Data availability

Data will be made available on request.

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